

Question 1 — Employee Management System (DDL + Constraints)

Create a table called `employees` with the following requirements:

- `employee_id` should be the primary key.
- `email` must be unique.
- `salary` should never be less than 25000.
- `gender` should allow only M, F, or O.
- `joining_date` should default to current date.
- `department_id` should reference a `departments` table.
- Employee age must be greater than or equal to 18.
- Prevent null values for employee name.

After creating the table:

1. Insert 5 records.
 2. Try inserting:
 - duplicate email
 - negative salary
 - invalid gender
 3. Observe and explain the errors.
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Question 2 — Banking Transaction System (Advanced DDL + DML)

Create the following tables:

accounts

- `account_id` → Primary Key
- `account_number` → Unique
- `balance` → Cannot be negative
- `account_type` → Only SAVINGS, CURRENT
- `created_at` → Default current timestamp

transactions

- `transaction_id` → Primary Key
- `account_id` → Foreign Key
- `transaction_type` → Only `DEPOSIT`, `WITHDRAW`
- `amount` → Greater than 0
- `transaction_time` → Default current timestamp

Tasks:

1. Insert sample data.
2. Withdraw money from an account using `UPDATE`.
3. Prevent balance from becoming negative.
4. Delete an account and analyze foreign key dependency behavior.
5. Add a new column `branch_name` later using `ALTER TABLE`.

Question 3 — E-Commerce Database Design (Complex Constraints)

Create tables for:

customers

products

orders

order_items

Requirements:

- Use proper primary and foreign keys.
- Product price must be greater than 0.
- Stock quantity cannot be negative.
- Order status should allow only:
 - `PLACED`
 - `SHIPPED`
 - `DELIVERED`
 - `CANCELLED`
- Use composite primary key in `order_items`.
- Prevent duplicate product entries in a single order.

Tasks:

1. Insert realistic Indian sample data.
 2. Update stock after placing orders.
 3. Delete a product that is already mapped in `order_items`.
 4. Add a constraint later to ensure customer phone numbers are unique.
 5. Rename a column using `ALTER TABLE`.
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Question 4 — University Course Registration (Advanced DDL Operations)

Create tables:

students

courses

faculty

enrollments

Requirements:

- `student_email` should be unique.
- `course_fee` should be between 5000 and 50000.
- `semester` should accept only:
 - `SPRING`
 - `SUMMER`
 - `FALL`
- Use foreign keys appropriately.
- Enrollment date should default to current date.
- Use cascading delete between student and enrollment tables.

Tasks:

1. Insert sample data.
2. Drop a constraint and recreate it.
3. Disable a constraint temporarily (if supported by DBMS).
4. Truncate one table and compare it with `DELETE`.
5. Drop one table and explain dependency issues.

Question 5 — HR Analytics System (Scenario-Based Advanced DML)

A company maintains employee salary and performance information.

Create tables:

employees

performance_reviews

salary_history

Requirements:

- Salary must always be positive.
- Rating should be between 1 and 5.
- Bonus percentage should not exceed 30%.
- Use foreign keys between all related tables.
- Use proper defaults and constraints.

Perform the following DML operations:

1. Insert 10 employee records.
 2. Give a 15% salary hike to employees in the **Data Engineering** department.
 3. Delete employees who resigned before 2022.
 4. Update performance ratings based on conditions.
 5. Copy high-performing employees into a new table called **top_performers**.
 6. Rollback one incorrect update using transaction control commands.
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